

- 9 (a)(i) The average time taken for the initial number of nuclei (or activity) of a particular radioactive nuclide to reduce to half its original value. **B1**
B1
- (ii) Decay is not affected by external or environmental factors (such as pressure, temperature etc) **B1**

$$\begin{aligned} \lambda &= \frac{\ln 2}{t_{1/2}} \\ &= \frac{\ln 2}{432.2 \text{ yrs}} \end{aligned}$$

(iii) **A1**

$$\begin{aligned} &= \frac{\ln 2}{432.2 \times 365 \times 24 \times 3600 \text{ s}} \\ &= 5.086 \times 10^{-11} \text{ s}^{-1} \end{aligned}$$

Number of undecayed atoms,

$$N = \left(\frac{1.00 \text{ g}}{241 \text{ g mol}^{-1}} \right) (6.02 \times 10^{23} \text{ mol}^{-1}) = 2.4979 \times 10^{21} \quad \text{C1}$$

(iv) $A = \lambda N$

$$\begin{aligned} &= (5.086 \times 10^{-11} \text{ s}^{-1}) (2.4979 \times 10^{21}) \quad \text{C1} \\ &= 1.27 \times 10^{11} \text{ s}^{-1} \\ &= 1.27 \times 10^{11} \text{ Bq} \quad \text{A1} \end{aligned}$$

(v) Loss of mass

$$\begin{aligned} &= 241.0568229 - (237.0481673 + 4.0026032) \quad \text{C1} \\ &= 6.0236 \times 10^{-3} \text{ u} \end{aligned}$$

$$\begin{aligned}
 \text{Energy} &= (0.0060524 \text{ u} \times 1.66 \times 10^{-27} \times (3.00 \times 10^8)^2) & \text{C1} \\
 &= 9.0423 \times 10^{-13} \text{ J} \\
 &= 5.65 \text{ MeV} & \text{A1}
 \end{aligned}$$

- (b) For nuclei having high nucleon numbers, the binding energy per nucleon decreases with larger nucleon numbers. **B1**

When two such nuclei fuse together, they will produce a daughter nucleus which has an even larger nucleon number and smaller binding energy per nucleon. **B1**

The total binding energy of the products is less than that of the initial nuclei, hence there is an increase in the total mass of the system, and energy has to be supplied for such a reaction to take place. **B1**

- (c) (i) ${}^4_2\alpha$ **B1**

$${}^1_1\text{H} \quad \text{B1}$$

- (ii) 1. Initially alpha particle must have some kinetic energy. **B1**

2.

$$1.1 \text{ MeV} = 1.1 \times 1.6 \times 10^{-13} = 1.76 \times 10^{-13} \text{ J} \quad \text{C1}$$

$$E_k = \frac{1}{2}mv^2 \quad \text{C1}$$

$$1.76 \times 10^{-13} = \frac{1}{2}(4 \times 1.66 \times 10^{-27} \times v^2) \quad \text{C1}$$

$$v = 7.3 \times 10^6 \text{ m s}^{-1} \quad \text{A1}$$

~ THE END ~